

Betaherpesvirus Sequences in Eastern Spiny Mice and Wagner's Dipodils

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ABSTRACT: Eastern spiny mice (*Acomys dimidiatus*; also known as Sinai spiny mice) have been extensively studied in terms of the influence of parasite load on population size and reproductive fitness. The physical isolation of these rodent populations makes them interesting models for disease interactions in a real-life population as opposed to a laboratory. We identify betaherpesvirus sequences in eastern spiny mice and Wagner's dipodils (*Dipodillus dasyurus*), species that inhabit dry montane wadis (dry creek valleys) of the Sinai, highlighting the need for a comprehensive analysis of the full pathogen repertoire of these rodents in long-term studies.

Key words: *Acomys dimidiatus*, Betaherpesvirus, *Dipodillus dasyurus*, eastern spiny mice, herpesvirus, pan-herpesvirus PCR, rodents, Wagner's dipodils.

We characterize herpesviruses in a population of Egyptian rodents. Our study animals originated from St Katherine's Protectorate in Egypt. This area encompasses the Sinai Mountains, a unique range of montane habitats based around interconnected dry creek valleys (wadis). The region has been selected as a conservation park because of its high biodiversity. The rodent population in this region has been extensively studied in an ongoing long-term research program in relation to their parasite burden and the effects this has on breeding fitness and population dynamics (Behnke et al., 2000; Barnard et al., 2003; Behnke et al., 2004; Bajer et al., 2006). The eastern spiny mouse (*Acomys dimidiatus*) inhabits the Bedouin gardens (typically, but not exclusively, located in the bottoms of the wadis), whereas Wagner's dipodil (*Dipodillus dasyurus*)

generally lives higher up the wadi slopes, but the species have overlapping ranges within the wadis.

The Bedouin of the mountains of South Sinai build walled orchard gardens where they grow vegetables and cereals and fruit and nut trees. They are virtually unique in the Middle East for this practice, which derives from the Byzantine anchorites who settled in the area from the fourth century BC. Some gardens are therefore probably 1,500 yr old, but others were built within the last 20 yr. Originally built around productive wells, increasingly they can be sited elsewhere because plastic pipes or sand dams can bring or retain ephemeral water to irrigate the crops. The gardens boost and maintain biodiversity in the hyperarid landscape (Zalat and Gilbert, 2008), and many animals are found associated with them, such as the eastern spiny mouse.

Herpesviruses are well known for manipulating their host's immune responses (Mocarski et al., 2007) and it has been shown in laboratory studies that induction of Th1 immune cellular responses to viruses can inhibit the host's ability to mount the Th2 responses necessary for the control of helminth parasites (Fenton, 2008). Therefore, knowledge of both a host's parasite and viral loads is necessary for a complete picture of the effects of these microbes on population dynamics. This study used samples collected by researchers in an ongoing study on parasite burdens in eastern spiny mice to identify herpesviral sequences in spiny

mice and Wagner's dipodils living in this unique wadi environment.

Trapping sites included Wadi El Arbaein (38°33'N, 33°56'E), Wadi Gebal (28°32'N, 33°54'E), Wadi Gharaba (28°38'N, 33°53'E), and Wadi Tlah (28°33'N, 33°56'E). Animals were live trapped with the use of Sherman traps, and trapping primarily targeted eastern spiny mice living in and around the walls of the Bedouin gardens. Traps were inspected early in the morning and closed during the day to prevent animals entering because of the hot daytime temperatures. A proportion of the captured animals were taken to the field station, euthanized and subjected to postmortem examinations as described by Barnard et al. (2003). The animals were a random subset of 310 rodents live-sampled in August–September 2008, of which 79% were spiny mice, and 6% were dipodils. The remaining 15% were rodents from six other species. One hundred sixteen spiny mice were culled (under a permit from the park authorities) to allow recovery of intestinal helminths and the three dipodils were accidental deaths during anesthesia (this species being particularly susceptible to anesthesia).

At autopsy a 0.5-cm³ piece of liver was taken with sterile instruments and stored in RNAlater (Applied Biosystems, Austin, Texas, USA) at ambient temperature for 6 wk, then at –80°C until being processed. DNA was extracted from liver samples using the Nucleospin DNA Tissue kit (Machery Nagel, Düren, North Rhine-Westphalia, Germany) according to the manufacturer's protocol. This DNA was then subjected to nested polymerase chain reaction (PCR) with degenerate primers targeting the herpesvirus DNA polymerase gene (Ehlers et al., 1999). Second-round PCR products were visualized on a 1% w/v agarose gel stained with ethidium bromide. PCR products of the expected size were excised from the gel and purified with the use of the Nucleospin extract II kit (Machery Nagel), according to the manufacturer's protocol.

DNA from purified PCR products was cloned, with the use of the pGEM-T Easy Vector System (Promega, Southampton, Hampshire, United Kingdom), following the manufacturer's protocol. Plasmids were transfected into chemically competent *Escherichia coli* C2988J cells (NEB, Ipswich, Massachusetts, USA), which were cultured on nutrient broth/ampicillin agar plates. Plasmids were extracted with the use of the Nucleospin Plasmid kit (Machery-Nagel), following manufacturer's instructions, and plasmids were screened for herpesvirus insertion, with the use of the second-round herpesvirus PCR described above.

Of 89 samples from spiny mice and three from dipodils, four (two from each species) yielded appropriate-sized PCR bands (approximately 250 base pairs) and were further analyzed. Both dipodils were trapped in the Wadi Gebal, at 1,888 m, the highest of the four study sites. One infected Eastern spiny mouse was trapped in Wadi El Arbaein and the other in Wadi Tlah (see Behnke et al., 2004 for details of geographic locations).

Polymerase-chain-reaction products or plasmids were sequenced with the use of Big Dye terminator 3.1 (Applied Biosystems) chemistry and ethanol precipitation according to manufacturer's recommendations and an ABI Prism 3130 Fluorescent DNA Analyzer (Applied Biosystems). The nucleotide sequences from eastern spiny mice were identical and have been tentatively named *Acomid herpesvirus* 1 (AcHV1). Similarly, the nucleotide sequences from dipodils were identical to each other and have been designated *Dipodillid herpesvirus* 1 (DiHV1). Sequences have been deposited in the GenBank database under accession numbers HQ587046 (AcHV1) and HQ587047 (DiHV1).

These sequences were screened against the GenBank nucleotide database with the use of the megablast search function and showed similarities to several strains of *murid herpesvirus* (MuHV1, murine cytomegalovirus [MCMV]), the type species in the genus

Muromegalovirus (Davison et al., 2009). When translated both AcHV1 and DiHV1 were identical at the amino acid level. Nucleotide similarity between AcHV1 and DiHV1 was 99.6% and nucleotide similarity between these isolates and MuHV1 was 95.2%. Amino acid similarity between AcHV1 and MuHV1 was 99.1%.

The amino acid sequence of AcHV1 was aligned with representative betaherpesvirus sequences with the use of ClustalW. A phylogenetic tree was generated by the maximum likelihood method (PROML) in the PHYLIP package. Bootstrap values were obtained with the use of 100 replicates. The sequence clusters with sequences from other strains in the muromegalovirus genus (Fig. 1), including *Apodemus flavicollis* cytomegalovirus 1 (AflaCMV1), Rat cytomegalovirus (RCMV, murid herpes virus 2, MuHV2) and *Bandicota indica* cytomegalovirus 4 (BindCMV4; Ehlers et al., 2007).

We detected few PCR-positive animals in this population (2% of the eastern spiny mice). The small number of Wagner's dipodids in this survey makes prevalence calculations for betaherpesviral infection unreliable. Antibody prevalence of MuHV1 in wild house mouse (*Mus musculus*) populations may be as high as 65%, and that of human cytomegalovirus (human herpesvirus 5, HHV5) in humans 52–99% (Mocarski et al., 2007). However, these viruses have a long latent phase in their replication cycles during which viral antigen or DNA may not be detectable. PCR-based studies of herpesviral prevalence in rodents typically demonstrate much lower prevalence. For example, 5% of house mice from the United Kingdom were positive for the gamma herpesvirus MmusRHV 1 (Ehlers et al., 2007). It is also possible that liver is not a tissue in which these viruses replicate. For example, the primary site of replication for MuHV1 is in the salivary glands (Krmptotic et al., 2003).

The sequences we describe are unusually similar. Herpesviruses are normally

very host specific, and unrelated species would not normally have such similar herpesviruses (Davison, 2002). This may suggest a recent cross-species transmission event. However, only a very small portion of the DNA polymerase gene was sequenced for this study. Analysis of a larger portion of the virus genome would yield a more accurate representation of viral diversity.

Without a more detailed study, it is difficult to assess the impact of these herpesviruses on this rodent population. Cytomegalovirus infections in human, house mouse, and rat (*Rattus norvegicus* and *Rattus rattus*) populations are largely subclinical (Krmptotic et al., 2003; Mocarski et al., 2007). However, a carrier state of infection exists and immunosuppression of the host may result in recrudescence of viral replication and a variety of disease syndromes. For MCMV infection in house mice these disease syndromes may include liver damage, immunosuppression, pneumonitis, adrenal necrosis, myocarditis and bone marrow aplasia. Fetal and neonatal infection results in central nervous system replication and brain damage (Krmptotic et al., 2003).

Murine cytomegalovirus infection also has a number of detrimental subclinical effects on the host. The virus can cause reduced litter size and increased fetal abnormalities in laboratory mice (*M. musculus*) and experimental infection with MCMV increases susceptibility to a number of bacterial pathogens (Committee on Infectious Diseases of Mice and Rats, 1991). If the herpesviruses identified in eastern spiny mice demonstrate a similar pathology they will have an impact on reproductive status and disease resistance (including resistance to bacterial and parasitic pathogens) of these populations.

The identification of potentially immunomodulatory viral pathogens in these well-studied populations of rodents means that the potential for viral interference must be considered in studies examining their parasite load and reproductive fitness

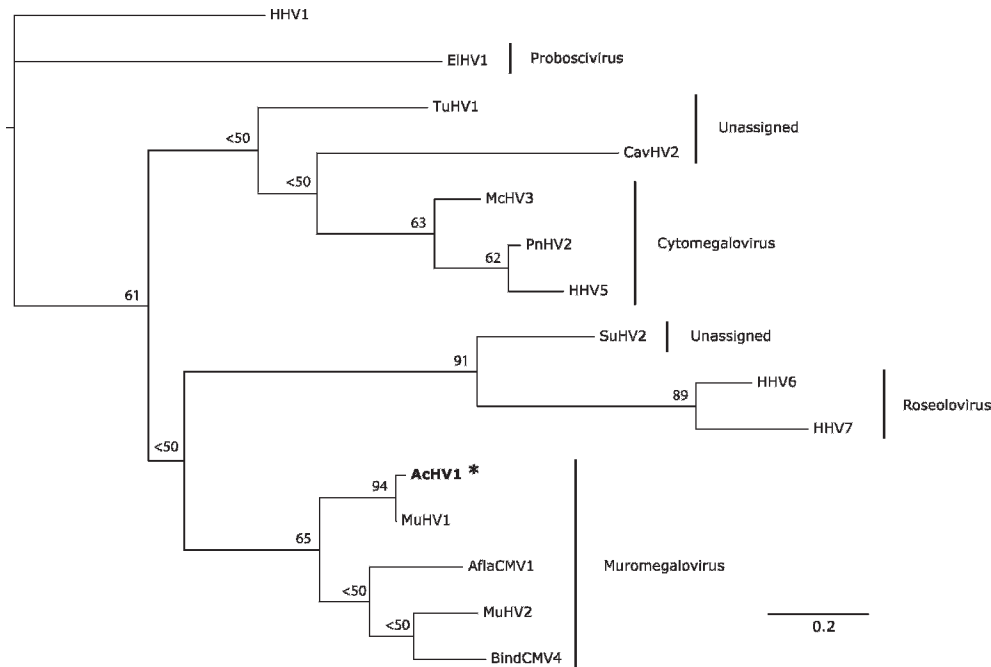


FIGURE 1. Maximum-likelihood phylogenetic tree constructed with the use of amino acid sequences encoded by the DNA polymerase gene of representative betaherpesviruses. Percent bootstrap values (based on 100 replicates) are indicated. Virus acronyms are followed by full names and accession numbers. As the AcHV1 and DiHV1 sequences were identical at the amino acid level, only the AcHV1 sequence is shown (marked with an asterisk). HHV 1 (*Human herpesvirus 1*, NC_001806), an alpha herpesvirus, was used as the outgroup; EIHV 1 (*Elephantid herpesvirus 1*, AF322977), TuHV 1 (*Tupaiaid herpes virus 1*, NC_002794), CaHV 2 (*Caviid herpesvirus 2*, FJ355434), McHV 3 (*Macacine herpesvirus 3*, NC_006150), PnHV 2 (*Panine herpesvirus 2*, NC_003521), HHV 5 (*Human herpesvirus 5*, NC_006273), SuHV 2 (*Suid herpesvirus 2*, AF268039), HHV 6 (*Human herpesvirus 6*, NC_001664), HHV 7 (*Human herpesvirus 7*, NC_001716), AcHV1 (*Acomys dimidiatus herpesvirus 1*, HQ587046) MuHV 1 (*Murid herpesvirus 1*, M73549), AflaCMV 1 (*Apodemus flavicollis cytomegalovirus 1*, EF125062), MuHV 2 (*Murid herpesvirus 2*, NC_002512), BindCMV 4 (*Bandicota indica cytomegalovirus 4*, EF125068).

(Fenton, 2008). The interactions between these herpesviruses and their host's parasite load and the effects this has on the population size and reproductive fitness of the host will be fertile areas for longer-term study.

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